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What is an Integer Linear Program?

NP-Hardness of Integer Linear Programs

✔

Video: NP Hardness of Integer Linear Programming

15 min

✔

Quiz: Integer Linear Programming

Submitted

Tutorial: Solving Integer Linear Programs using Python/Pulp

Vertex Cover as Integer Linear Program

Linear Programming Approximations of Vertex Cover

Solving Integer Linear Programs using Branch and Bound Algorithm

Problem Set # 2

Integer Linear Programming

🎉 Congratulations! You passed!

Grade received 100%

Latest Submission Grade 100%

To pass 80% or higher

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Review Learning Objectives

✔ Submit your assignment

Due Feb 18, 11:59 PM IST

✔ Receive grade

To Pass 80% or higher

1. Consider the integer linear programming problem:

$$\begin{aligned} \max \quad & 2x_1 + 3x_2 \\ \text{s.t.} \quad & x_1 \geq 0 \\ & x_1 \leq 9.8 \\ & x_2 \geq 0 \\ & x_2 \leq 10.5 \\ & x_1, x_2 \in \mathbb{Z} \end{aligned}$$

3 / 3 points

Try again

Select all true facts about this problem below, ensure no wrong facts are selected.

✔ The problem has an optimal solution and it is given by $x_1 = 9, x_2 = 10$.

✔ Correct
Correct, note that x_1, x_2 are integer valued variables.

✔ The LP relaxation achieves an optimal objective value of 51.1 whereas the ILP optimal solution is 48.

✔ Correct
Correct: $2 \cdot 9.8 + 3 \cdot 10.5 = 51.1$

□ The problem has an optimal solution and it is given by $x_1 = 9.8, x_2 = 10.5$.

✔ The integrality gap for this problem is 51.1/48

✔ Correct
Correct: integrality gap just like approximation ratio is always taken to be a number more than 1.

□ The integrality gap for this problem is 48/51.1

2. Select all the true facts about integer linear programming problems in relation to their LP relaxations. Make sure that no incorrect fact is selected.

3 / 3 points

□ If the LP relaxation is infeasible, it is possible that the ILP is feasible.

✔ If the ILP is unbounded, the LP relaxation is unbounded too.

✔ Correct
This is correct and follows directly from the fact that the feasible region of LP over approximates that of the ILP.

□ If the ILP is infeasible, the LP relaxation is also infeasible.

✔ If the problem involves maximizing an objective, the optimal ILP objective is always less than or equal to that of the LP relaxation.

✔ Correct
Correct because feasible region of ILP is a subset of that of the LP relaxation since we drop the integrality constraints.

3. The following questions concern the NP-completeness of integer linear programming. Please select all correct facts.

3 / 3 points

✔ The best known algorithms for solving ILPs take time that is exponential in the number of variables and constraints of the problem.

✔ Correct
Correct: otherwise, we would have resolved the P vs NP question

✔ If $P = NP$, we can solve ILPs efficiently in polynomial time.

✔ Correct
Indeed: that is what NP completeness of ILPs imply.

✔ There are ILPs whose LP relaxations yield integer solutions and thus can be solved using a linear programming solver.

✔ Correct
Correct but note that this is not a general algorithm. We could try a LP solver on the LP relaxation and if it works we succeed/ if it fails, we have to do something else. This is a correct statement but it is not a general algorithm, as stated.

✔ Every problem in NP can be reduced in polynomial time to an instance of ILP, and thus ILP solvers can solve any problem in NP.

✔ Correct
Exactly, this is what NP hardness means.

✔ Every Boolean SAT problem can be solved by reducing it to an ILP involving just binary variables.

✔ Correct
Correct: we showed this in our video.

4. Suppose we are given an ILP which seeks to minimize an objective functions subject to constraints. We solve the LP relaxation and find an optimal solution with the objective evaluating to 54.3

5 / 5 points

What can you say about the original ILP.

✔ If the ILP is feasible, its optimal solution must be greater than or equal to 54.3

✔ Correct
Correct: it is a minimization problem. Therefore, the ILP objective must be greater than or equal to that of the LP relaxation.

□ If the ILP is feasible, its optimal solution must be less than or equal to 54.3

✔ If the integrality gap of the ILP problem was known to be 3, then the ILP optimal solution will be in the range $[54.3, 3 \times 54.3]$

✔ Correct
Correct

✔ It is possible that the ILP's optimal solution (if it exists) will be 76.5×10^{231} .

✔ Correct
Sure: there is no general bound on the integrality gap.

✔ If the ILP is feasible, it is guaranteed to be bounded.

✔ Correct
Yes, this immediately follows from the fact that the ILP feasible region is a subset of that of the LP.

□ It is guaranteed to be feasible.

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